Double large primes on the genus-5 cover: half the candidates, twice the linear algebra

Study `research/humbert_genus5_cover_20261021/dlp_20261113` · 2026-11-13 · follows `lp_20261027` (PR #2710)

Scope. Toy fields only (q prime, $4001 \le q \le 16001$), the iota-folded index calculus on the descended genus-5 curve C (PR #2580). One campaign under `PROTOCOL.md`, frozen before data. Grid: 4 values of q × 5 seeds × 5 arms = 100 runs, same instance per (q, seed) in every arm, arm order rotated per seed, one frozen binary (`bin/humbert-v3`, sha256 prefix `a8ff53874ff869a9`). Rho runs once per instance; every arm is compared with that instance's expected-steps rho $\sqrt{\pi r/2}$ × measured s/step.

Host and standing. 4-vCPU shared cloud VM, no isolation (L0); isolab-eks (AGENTS.md §9a) was not reachable from the session. Every wall-time number below is **exploratory** and an **unverified assertion**; no row goes to the IC leaderboard (no replay certificate, validator report or `VALIDATION.json`). The candidate and mulmod counts are host-independent.

Arms. `pre`: `iota` with the linear-algebra precheck (the PROTOCOL_v2 baseline). `pre_aa`: the same again in its own process (A/A). `lp50`: single large prime, small base of half the x-values. `dlp50`, `dlp35`: double large prime with small bases of 50 % and 35 %. A residual with at most two points outside the small base is reduced against stored pivots, one per large prime (`cur·b − pivot·a` cancels that large prime); a row that reaches zero large primes is a full relation; a row left with one or two unpivoted large primes becomes the pivot for one of them; rows with more are dropped. Large-prime logs follow from the pivots after the linear algebra.

Correctness. All 100 runs solved the DLP and passed the pushforward ($8P$), $T(P) \neq 0$, iota-pair and $X_\infty = 4c$ checks (`humbert summarize-dlp`: "all correctness gates pass: true").

1. Predictions against the data

q	A/A median [min, max]	pre/dlp50 median [min, max]	dlp50 faster	lp50/dlp50 median	cand dlp50/lp50 max	IC/rho-exp median: pre, lp50, dlp50, dlp35
4001	1.011 [0.983, 1.027]	1.171 [1.149, 1.567]	5/5	1.711	0.5113	1.563, 2.063, 0.998, 1.284
8009	0.992 [0.968, 1.082]	1.148 [1.019, 1.328]	5/5	1.239	0.5119	2.344, 2.755, 2.180, 2.317
12007	0.990 [0.949, 1.024]	1.148 [1.107, 1.366]	5/5	1.084	0.5117	3.150, 3.190, 2.846, 2.835
16001	0.991 [0.971, 0.998]	1.406 [1.174, 1.952]	5/5	0.967	0.5126	2.280, 1.673, 1.650, 1.594

Table 1: Paired ratios on the same instance, 5 seeds per q (`results/summary_v3.csv`, from `campaign_a` and `campaign_b`). Above 1, dlp50 is faster.

- **R1 (counted): holds.** dlp50 needs fewer relation candidates than lp50 on all 20 instances; the ratio is 0.472–0.513, flat in q.
- **R2 (wall time): holds.** dlp50 is faster than `pre` on 5/5 seeds at each of $q = 8009$, 12007 and 16001 (medians 1.15, 1.15, 1.41). The A/A spread is within ±5 % except one instance at 8.2 %.
- **R3: holds.** Median IC / expected-steps rho for dlp50 is below `pre` at each $q \ge 8009$: 2.18 vs 2.34, 2.85 vs 3.15, 1.65 vs 2.28. Rho is still faster at every q for every arm; that is the distance remaining.
- **R4 (exploratory): no exponent gain.** The log–log slope of median t_{ic} from 4001 to 16001 is 1.68 for dlp50 and 1.77 for `pre`. Both are far above the $q^{1.6}$ that GTTD predicts asymptotically for genus 5.

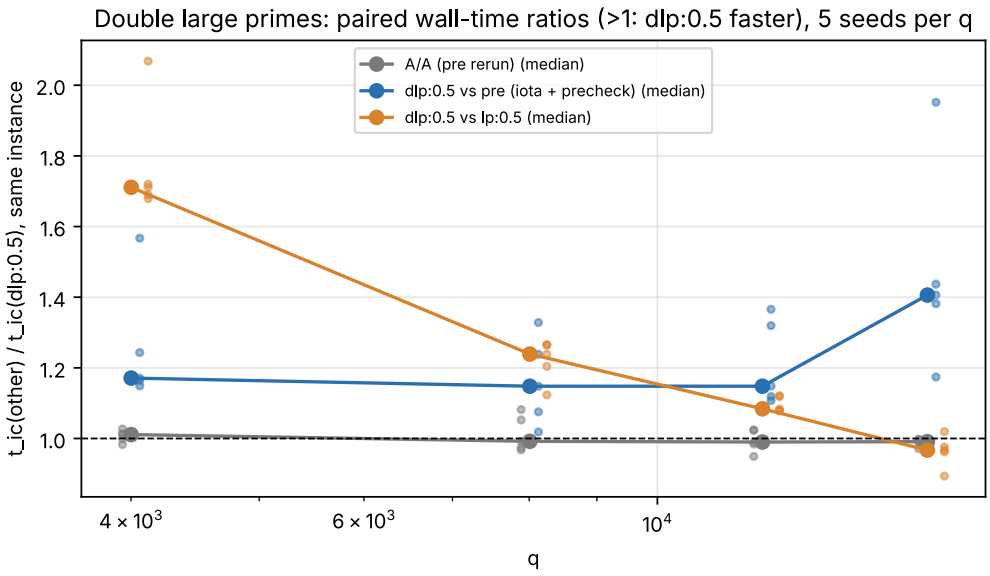


Figure 1: Paired wall-time ratios; points are instances, lines are medians.

2. The unpredicted result: dlp's advantage over lp vanishes with q

The median of lp50/dlp50 falls from 1.71 at $q = 4001$ to 1.24, 1.08 and 0.97. At $q = 16001$ dlp50 is no faster than single large primes, even though it still needs half the candidates. The counted phase split explains it:

q	arm	t_rel (s)	t_LA (s)	t_descent (s)	candidates	LA mulmods	unknowns
4001	pre	2.6505	3.0056	0.2069	50061	426843228	1987
4001	lp50	7.2284	0.8865	0.3965	137157	242162882	994
4001	dlp50	3.5963	1.0646	0.2792	70085	293862299	994
4001	dlp35	5.5333	0.3786	0.5506	107631	112664905	697
8009	pre	6.2354	12.6042	0.2593	106862	1794822300	4018
8009	lp50	15.6406	4.3471	0.5847	268542	623134282	2010
8009	dlp50	7.7782	8.9686	0.2651	135355	1320052626	2010
8009	dlp35	12.2322	6.0023	0.3421	206813	919338852	1407
12007	pre	9.8536	28.4302	0.2609	147287	3951654408	6032
12007	lp50	26.0492	10.0313	0.3494	413911	1399602606	3017
12007	dlp50	12.4291	20.4252	0.3072	201146	2955177696	3017
12007	dlp35	19.0086	13.8083	0.4434	311291	2076715944	2112
16001	pre	13.9581	57.1626	0.2811	223486	8112207510	7987
16001	lp50	33.0839	16.7972	0.4120	541407	2444846648	3994
16001	dlp50	16.4960	34.7373	0.2986	270502	5208051842	3994
16001	dlp35	25.2939	24.6373	0.2984	414004	3701046542	2797

Table 2: Medians over the 5 instances per q (`results/phases_v3.csv`).

At every q, dlp50 and lp50 solve the same number of unknowns. dlp50 needs 0.50× the candidates but does 1.21×, 2.12×, 2.11× and 2.13× the LA mulmods. Wiedemann work at a fixed dimension is proportional to the matrix weight, so dlp's reduced rows are about twice as heavy. Each reduction step adds a pivot's whole support to the row, and chains of pivots compound that fill-in. Relation collection grows more slowly in q than the linear algebra does (`pre`: t_{rel} 2.6 ⟶ 14 s, t_{LA} 3.0 ⟶ 57 s). The candidate saving is therefore paid for in the phase that dominates at larger q. dlp35 shrinks the matrix (fewer unknowns), not its density, and is the fastest arm at $q = 16001$ by median IC/rho (1.59). The margin over lp50 and dlp50 is inside the instance spread.

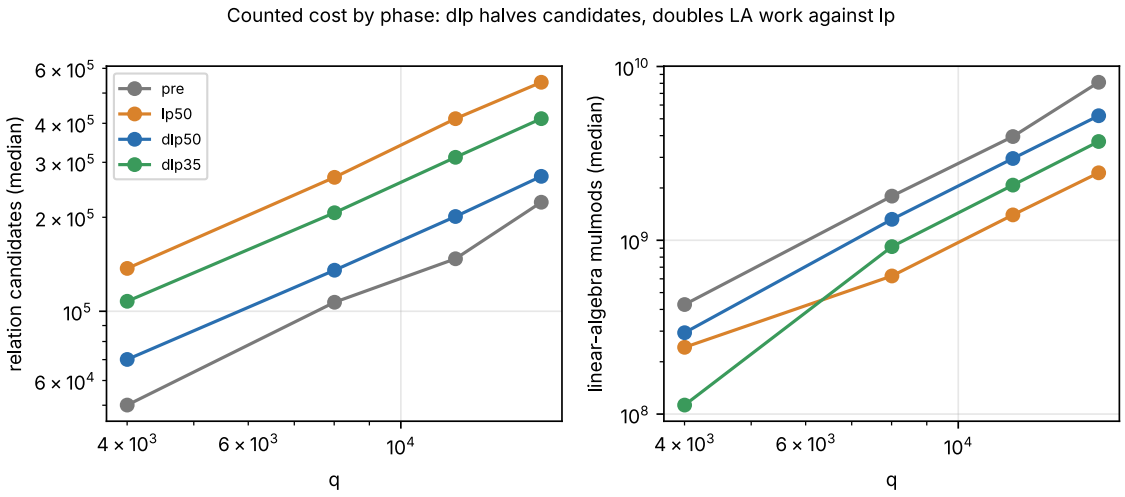


Figure 2: Counted cost per phase: candidates (left) and LA mulmods (right), log–log.

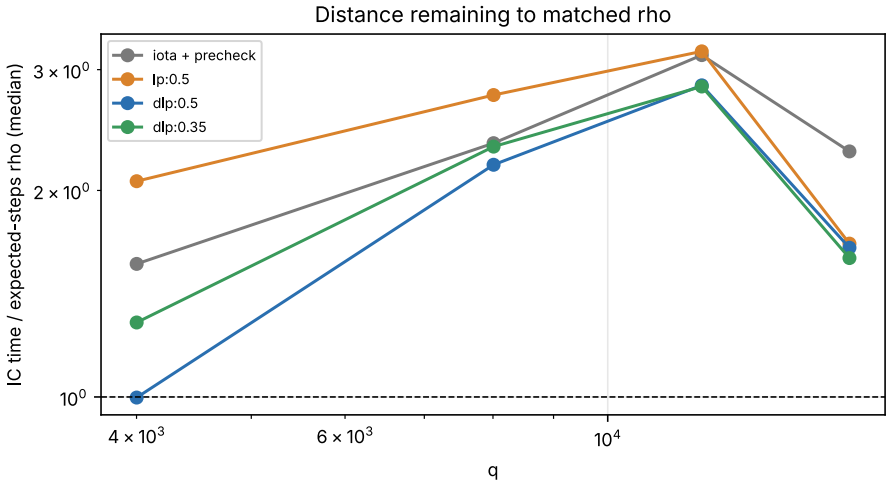


Figure 3: Median IC time over expected-steps rho; 1 is parity.

3. What this means for the lead

- Against the fixed baseline (`pre`), double large primes are a measured improvement at every q tested (R2, R3). This is an unverified, unisolated (L0) assertion, not a leaderboard row.
- Against single large primes, they are not, from $q \approx 16000$ on. The best arm per q at the top of the grid is any of lp50, dlp50 or dlp35, all within noise of each other.
- The exponent did not move at toy scale (R4). The binding constraint is linear-algebra density, not the relation yield.
- **Next step (proposed, not run):** a weight-bounded reduction. Drop or defer any reduced row whose support exceeds a bound w, or cap the pivot-chain length. Trade back some of the 2× candidate saving for the 2× LA weight. Prediction to freeze: at $q = 16001$, LA mulmods drop to within 1.2× of lp50 while candidates stay at most 0.75× lp50.

Labelling caveat. The `dlp_cycles` field counts every full row emitted by the reducer, including rows that had no large prime from the start; it is not a count of graph cycles. No analysis uses it.

Reproduce. `BIN=dlp_20261113/bin/humbert-v3 BIN_SHA256_PREFIX=a8ff53874ff869a9 PRIMES="…" ./run_v3.sh <new-dir>; humbert summarize-dlp dlp_20261113/results campaign_a/results/v3.jsonl campaign_b/results/v3.jsonl; python3 plot_dlp.py`. Bench record: `benchmarks/records/20261113T1559Z-humbert-genus5-dlp.json`.